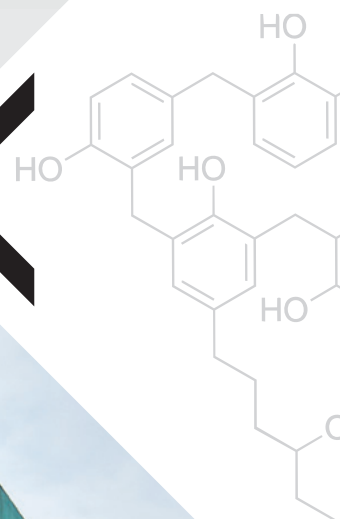




Kanoria Chemicals
& Industries Limited



Responsible Care®
OUR COMMITMENT TO SUSTAINABILITY



Chemistry of Values and Value Additions

PHENOL FORMALDEHYDE RESINS



Founded in 1960, Kanoria Chemicals & Industries Ltd. (KCI) is a leading manufacturer of chemical intermediates and specialties in India, with a focus on products for infrastructure and construction. KCI has strong capability with respect to wealth creation for all its stakeholders while adopting sound corporate governance practices. The company steadfastly believes in the core values of sustainability, transparency, ethical business practices and maintaining high standards of corporate governance. We are well recognized through awards in the areas of environment management, resource use efficiency and social responsibility.

Kanoria Chemicals and Industries Ltd. (KCI) operates three alco-chemical manufacturing facilities in India, producing a range of products, located at Ankleshwar (Gujarat), Visakhapatnam (Andhra Pradesh) and Naidupeta (Andhra Pradesh). We manufacture Phenol Formaldehyde resins at Ankleshwar and Visakhapatnam.

KCI's manufacturing facilities can produce a wide range of Novolac, Resole and similar resins. They feature extensive automation for precise mass transfer and temperature control, leading to products of high quality and consistency under safe conditions. The resin powdering facilities are virtually hands-free and dust-free. With proximity to all major chemicals, our plants can cater to the regular, special and urgent needs of a wide range of Phenol Formaldehyde resin consumers.



OUR STRENGTHS

LOCATION

Pan-India coverage and proximity to ports



RELIABILITY

Critical RMS (Formaldehyde, Hexamine) being produced in-house



MANUFACTURING EXCELLENCE

World class facilities with strong technical support



SYSTEMS

Certified for ISO and RC systems



QUALITY & CONSISTENCY

Highly automated, statistical process control



ANALYTICS

Use of modern techniques during product development



KCI'S PHENOLIC RESIN PRODUCT RANGE

Novolak Resin Powders

KCI –Unmodified Resins For Friction Materials

Grade	Flow	Hexamine Content (%)	Melting Point	Applications / Features
KRC2012P	Medium	8.0 – 10.00	90 – 100	High free phenol content resin maximum - 5% below
KRR2004P	Medium	8.5 – 9.5	85 – 95	Low free phenol containing resin – environment friendly
New Grade	low	9.5 – 10.5	85 – 95	Ultra low free phenol maximum -0.5% below
KRR2006P	High	9.5 – 10.5	82 – 90	Brake pads in friction industry
KRR2002P	Short	9.5 – 10.5	90 – 100	good strength and workability
KRR2001P	High	11.0 – 13.0	85 – 90	Good mechanical strength, high flame reterdancy
New Grade	Low	10.0 – 12.0	85 – 95	Reactivity -proprietary
New Grade	Medium	9.0 – 10.0	80 – 90	Reactivity -proprietary
KRR2003P	Medium	9.0 – 11.00	80 – 90	Low porosity and high density
KRR2005P	Medium	9.0 – 10.0	87 – 92	Friction industry

KCI –Modified Resins For Friction Materials

Grade	Flow	Hexamine Content (%)	Melting Point	Applications / Features
KRR2047P	Very Low	9.5 – 10.5	90 – 95	NBR modified for friction industry (10% modified)
New Grade	Low	8.5 – 9.0	85 – 95	NBR modified for friction industry (18% modified)
KRR2038P	Medium	9.0 – 11.0	78 – 84	CNSL base resin for friction
New Grade	Long Flow	9.0 – 11.0	74 – 80	CNSL base resin for friction
KRR2040P	Medium	9.0 – 11.0	79 – 85	Cardanol base resin for friction
New Grade	Low	10.0 – 12.0	80 – 90	Cardanol base resin for friction
KRR2021P	Medium	8.5 – 9.5	85 – 90	Fast cure for brake lining and friction components
KRR2027P	Medium	8.5 – 9.5	80 – 90	High thermal stability and mechanical strength
KRR2023P	High	11.0 – 12.5	86 – 92	High wear resistance
KRR2026P	Medium	8.0 – 9.0	80 – 90	Ultra-low free phenol and dust-suppressed resin

For Refractories

Grade	Flow	Hexamine Content (%)	Melting Point	Applications / Features
KRR2002P	Short	9.5 – 10.5	90 – 100	Mag carbon bricks; for good green strength and workability
KRR2003P	Medium	9.0 – 11.00	80 – 90	Tundish Nozzles and slide gates; for low porosity and high density

For Abrasives

Grade	Flow	Hexamine Content (%)	Melting Point	Applications / Features
KRR2002P	Medium	8.5 – 9.5	85 – 95	High strength and flexibility
KRR2004P	Short	7.0 – 8.0	85 – 95	High grinding ratio for bonded abrasive
KRR2026P	Medium	8.0 – 9.0	80 – 90	Ultra-low free phenol and dust-suppressed resin

For Moulding Materials

Grade	Flow	Hexamine Content (%)	Melting Point	Applications / Features
KRR2001P	Medium	11.0 – 13.0	85 – 90	Good mechanical strength, high flame reterdancy

For Nonwoven felt

Grade	Flow	Hexamine Content (%)	Melting Point	Applications / Features
KRR2003P	Medium	9.0 – 11.0	80 – 90	Felt resin for nonwoven product
KRR2045P	Medium	10.0 – 11.0	80 – 90	Fire Retardant Resin

For Sleeves

Grade	Flow	Hexamine Content (%)	Melting Point	Applications / Features
KRR2005P	Medium	9.0 – 10.0	87 – 92	Foundry sleeves and tundish boards

For Refractory's

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRR4003N	8500 – 9500@25°C	65 – 70	Mag carbon bricks and silica bricks
KRR4004N	10000 – 13000@20°C	68 – 70	Slide gates, nozzles and manoblock stopper
KRR4005N	6500 – 7500@25°C	68 – 72	Tap hole clay , Mag carbon bricks and mudgun mass
KRR4006N	12700 – 13700@25°C	75 – 78	Mag carbon bricks and AG products
KRR4009N	7000 – 9000@30°C	75 – 78	Mag carbon bricks, ASC products
KRR4011N	800 – 900@40°C	60 – 64	Tap hole clay , Mag carbon bricks and Flow control product

For Foundry

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRF4001N	100-130sec@30°B4cup	70 – 73	Shell moulding cores/moulds

Resole Resins

For Refractories

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRR3005R	1500 – 2200 cps	75 – 78	Tap hole mass – for high strength and flexibility
KRR3017R	3000 – 4000 cps	80 – 82	Mag carbon bricks – for different workability
KRR3019R	6000 – 7000 cps	82 – 84	Mag carbon bricks – for high strength with good work life

For Bonded Abrasive

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRA3007R	45-55s@25°C , B4cup	60 – 65	Bonded abrasive , for excellent grain wetting

For Composites

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRC3013R	15-20s@25°C , B4cup	55 – 60	Very low formaldehyde content impregnating resin
KRR-3002R	1000 – 3000cps	65 – 75	Fibre Reinforced Composites Resins
KRR3031R	250 – 350 cps	70 – 75	Rail wheel composites

For Friction

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRC-3008R	50 – 150cps	55 – 60	Additive Resole Resin for Friction

For Coated Abrasive

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRR-3028R	7000 – 10000 cps	75 – 80	Lignin modified Resole Resin

For Abrasive

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRC-3004R	200 – 400 cps	65 – 70	Coated Abrasive Resins

For Foundry Specialities

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRF3006R	29-33s@30°C ,B4cup	70 – 74	Acid cure,Low gas emission, long bench life range, nitrogen free
KRF3012R	24-27s @30°C ,B4cup	48 – 52	Ester cure, Environment friendly, excellent sand reclaimability, long shelf life

For Melamine Formaldehyde Resins

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KRR3030MF	16-20s @30°C ,B4cup	50 – 54	Laminates

For Urea Formaldehyde Resins

Grade	Viscosity @ 25°C	Solid Content (%)	Applications / Features
KUF1001	16-20s @30°C ,B4cup	42 – 44	Laminates

What makes us different



Expert
**Technical
Support**



Customized
**Product
Development**
based on
applications
and processes



Differentiated products
for Present Needs and
**Innovative
Research**
for future needs

Corporate Group

KCI is a part of a diversified, well-established corporate family. We are a values-driven group. We believe in a focus on quality, partnerships, and social responsibility.

The group's interests can be broadly divided into three verticals: chemicals, electronics and textiles.

CHEMICALS		ELECTRONICS	TEXTILES	TECHNOLOGY
 Kanorja Chemicals & Industries Limited KCI - Manufacturing of chemical intermediates and specialties. Headquartered in New Delhi, with manufacturing 3 facilities in West and South India.		 APAGCoSyst - Design and manufacturing of automotive concept lighting and comfort electronics; electronic manufacturing services. Headquartered in Switzerland, with manufacturing facilities in Czech Republic and Canada.	 Kanoria Africa Textiles - Denim fabric and garment manufacturing in Ethiopia.	 Intrust Global E-Services - Data science and engineering, software solutions and process outsourcing for the legal and finance industries. Headquartered in Bengaluru.
 KPL INTERNATIONAL LTD. <i>chemistry of partnerships</i> KPL - Marketing of quality products in India, mainly in the field of chemicals, polymers, paper & paper chemicals, sourced from globally acclaimed manufacturers. Headquartered in New Delhi.				



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